HACKATHON // USAGE-BASED INSURANCE

BimaRide.AI

Smart Ride Insurance Score

Turning every kilometre into a risk signal — telematics, AI, and insurance rewards on one platform. Price the ride, not just the rider.

642 km ridden | 3 harsh brakes | 1 overspeed | 0 crashes

Team <YOUR TEAM NAME> // Track: InsurTech & AI // <EVENT DATE>



Motorcycle insurance is flat, blind, and unfair.

TODAY

Insurers price on static proxies

The same premium for a 21-year-old speed-lover and a 40-year-old veteran with identical bikes.

- Age, city, bike model, IDV, NCB — that's it.
- Zero visibility into how the person actually rides.
- Rash and safe riders pay the same rupee.
- No feedback loop to change behaviour.
- Claims are reactive — after the crash.
- 70%+ of Indian 2W accidents are behaviour-driven*.

WITH BIMARIDE AI

Price the ride, not just the rider

A living, per-trip Rider Safety Score drives premium, rewards, and coaching.

- Live telematics from every smartphone.
- Behaviour + road context = true risk.
- Rider Safety Score 0–100, updated per trip.
- Safer riders unlock rewards + renewal benefits.
- Insurer sees portfolio risk in real time.
- Emergency SOS — crash detection saves lives.

One platform, three surfaces.

B2C

Rider App

Flutter, cross-platform, 90-second onboard.

- Live trip tracking + score card
- AI coaching in plain language
- Emergency SOS (auto-crash)
- Rewards, streaks & leaderboards

CORE

AI Risk Engine

Sensor fusion → ML → explanations.

- Multi-sensor event detection
- XGBoost risk model + SHAP
- LLM-grounded rider coaching
- Road-context risk enrichment

B2B

Insurer Console

Next.js dashboard for underwriting & ops.

- Portfolio risk in real time
- Rider segmentation & cohorts
- City risk heatmap
- Reward & renewal engine

One data pipeline. One score. Value on both sides of the policy.

Rider gets safer & cheaper. Insurer prices smarter. Everybody wins.

SYSTEM ARCHITECTURE

Sensor → score → insurance action.

01 MOBILE EDGE
GPS · IMU
Gyro · Speed
On-device buffer

02 INGEST API
FastAPI async
JWT · Redis
Kafka bus

03 EVENT ENGINE
Windowing
Rule + DSP
Feature extract

04 ML RISK MODEL
XGBoost + SHAP
LLM explain
Score 0–100

05 INSURANCE ACTIONS
Rewards · Renewal
Alerts · Claims
Underwriting

DATA & STORAGE LAYER

PostgreSQL
Trips · Users · Policies

TimescaleDB
Sensor streams · 50 Hz IMU

Redis
Live score cache · sessions

S3 / object
Raw logs · ML training data

RIDER APP · FLUTTER

REST + WebSocket · on-device retry · trip replay · score card · SOS trigger · gamification.

INSURER CONSOLE · NEXT.JS

Portfolio KPIs · heatmaps · cohort filters · claims correlation · reward approvals · exports.

The smartphone is the sensor pack.

GPS

Speed, distance, route, time-of-day, dwell.

`1 Hz · ~5 m accuracy`

Accelerometer

Longitudinal + lateral g-forces, impact spikes.

`50 Hz · 3-axis`

Gyroscope

Yaw, roll, pitch — lean angle and cornering behaviour.

`50 Hz · 3-axis`

Magnetometer

Heading, direction changes, route disambiguation.

`10 Hz · compass`

Phone State

Screen on/off, calls during ride, orientation flip (crash cue).

`event-based`

Context APIs

Weather, road type, speed limits, accident zones (OSM + partners).

`external enrichment`

Zero extra hardware. No OBD, no dongle. Onboard in 90 seconds.

Every Android + iOS device becomes a certified telematics unit.

Raw signals → semantic ride events.

RAW STREAM · PER SECOND

```
# timestamp    speed         long-accel
t=14:32:10    v=72 km/h    ax=-0.4 g
t=14:32:11    v=63 km/h    ax=-0.9 g
t=14:32:12    v=51 km/h    ax=-1.1 g
t=14:32:13    v=41 km/h    ax=-0.8 g
t=14:32:14    v=35 km/h    ax=-0.3 g

delta_v              = -37 km/h in 4s
peak_decel           = 1.1 g
phone_orient_stable  = true
context              = urban · limit 50
```

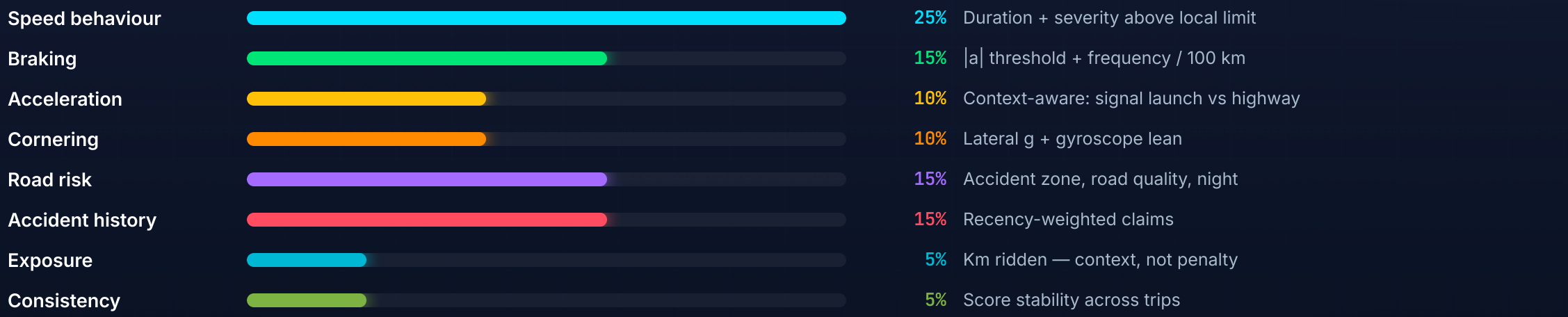
DERIVED EVENT

HARSH BRAKING

Severity	HIGH · |a| > 0.8 g
Duration	4.0 s
Context	Urban · wet weather
Score hit	−3 pts on Braking
Coach tip	Increase following gap

5 event classes detected: **harsh brake, harsh accel, aggressive cornering, overspeed, crash-suspect**.

Weighted, explainable, per-km normalised.




```
Score = 0.25·Speed + 0.15·Brake + 0.10·Accel + 0.10·Corner + 0.15·RoadRisk + 0.15·History + 0.05·Exposure + 0.05·Consistency
```

FAIR NORMALISATION

Per-100 km rates beat raw counts.

LOOKS SAFE — ISN'T

Rider A

DISTANCE
100 km / month

HARSH BRAKES
5 events

5.00
EVENTS / 100 KM

TRULY SAFE

Rider B

DISTANCE
2,000 km / month

HARSH BRAKES
5 events

0.25
EVENTS / 100 KM

WHY THIS MATTERS

Raw event counts punish high-mileage commuters and reward garage-queen bikes. BimaRide normalises every behavioural signal **per 100 km** before scoring. Result: the score reflects **how you ride**, not how often.

Same behaviour, different context, different risk.

ROAD RISK FACTORS

Accident history	25%	Zone-level claims density
Traffic density	20%	Live congestion overlay
Road quality	20%	Potholes, surface events
Speed violations	15%	Aggregate zone signal
Night riding	10%	Visibility + fatigue
Weather	10%	Rain, fog, temperature

RIDER A · LOW CONTEXT RISK

- Highway · well-lit · dry
- Low accident-density area
- Daytime commute
- Same behaviour → **lower risk**

RIDER B · HIGH CONTEXT RISK

- Urban · pothole-heavy road
- Historically accident-prone zone
- Night + rain
- Same behaviour → **higher risk**

ML for prediction, LLM for explanation.

MODEL 1 · RISK PROBABILITY

XGBoost Classifier

```
# Features (per trip)
avg_speed, max_speed, overspeed_seconds
harsh_brakes_per_100km
harsh_accel_per_100km
cornering_events, lateral_g_p95
night_ratio, weather_score
road_quality, claims_last_3y

# Target
y = P(claim next 90 days)

# Output
score       = 100 - 100·P(y)
shap_values = per-feature contribution
```

MODEL 2 · EXPLAIN + COACH

LLM (Grounded on SHAP)

Rider: Why did my score drop from 93 to 86?

AI: You had 8 harsh-braking events this week vs 2 last week, and spent 18% more time above the local limit — mostly on Ring Road. Fix these and your score returns to 92+ in ~10 days.

LLM outputs are grounded on SHAP contributions — **no hallucinations**, always cite the numeric driver.

The feature that saves lives, not just money.

SIGNAL	DETECT	CONFIRM	DISPATCH	SHARE
Speed 67 → 0 Decel > 3g Orient flip	Fusion model P(crash) > 0.85	"Are you OK?" 15 s countdown	SMS + call to emergency contacts	GPS + last speed to responder

WHY THIS WINS

- **Rider trust** — Insurance becomes a safety product, not a bill.
- **Faster claims** — Crash telemetry auto-generates a FIR-ready timeline.
- **Fraud control** — Sensor timeline validates or denies staged claims.
- **Regulatory fit** — Aligns with IRDAI's push for tech-led safety & UBI.

Turn every safe kilometre into a benefit.

90 – 100

Safe Rider

- Max reward points
- Priority renewal
- Partner discounts
- "Safe Rider" badge / NFT

75 – 89

Good Rider

- Moderate rewards
- Personalised coaching tips
- Small partner offers
- Score-improvement plan

60 – 74

Needs Improvement

- Guided coaching
- Weekly targets
- Micro-rewards on wins
- In-app challenges

< 60

High Risk

- Safety alerts
- Compulsory tutorial
- Human review flag
- Risk-mitigation plan

Pricing impact is illustrative — production premium is subject to IRDAI-approved actuarial models.

Behaviour change is a habit loop.

★

Top 8%

You beat 92% of riders in your city this month.

7

7-day streak

No harsh braking for seven days straight.

+12

Score climb

Score improved by 12 points in 30 days.

100

Century challenge

Complete 100 km with zero harsh events.

↻

Corner Master

50 clean corners in a single week.

◐

Night Owl Safe

10 safe rides after 9 PM.

Optional city leaderboards — opt-in, alias-based, GDPR/DPDP friendly.

Portfolio-grade telematics analytics.

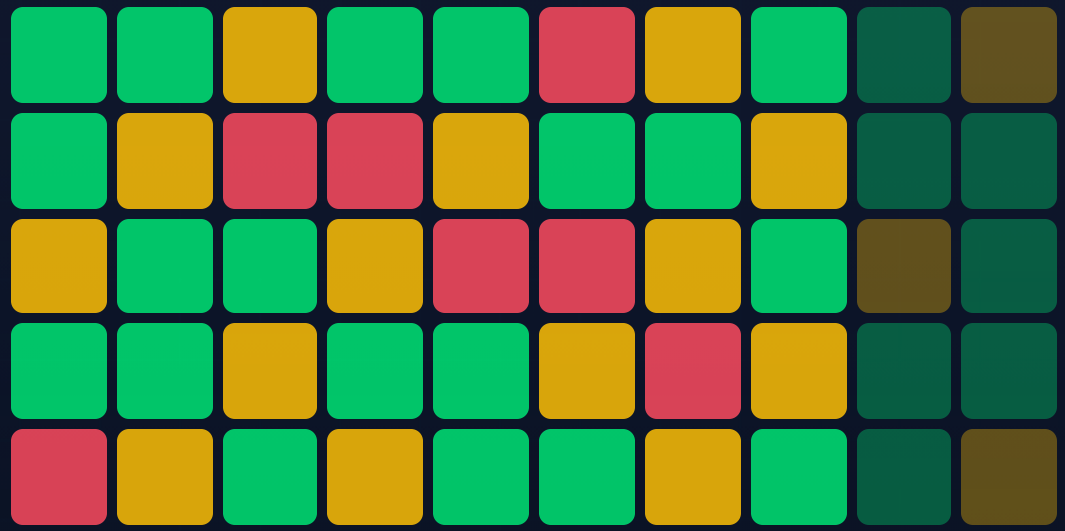
ACTIVE RIDERS	AVG SCORE	SAFE · MED · HIGH	CRASH ALERTS
48,921	81	62 / 28 / 10 %	143

RIDER DRILL-DOWN

RID-10231

Score	91
Distance	642 km
Harsh braking	3
Overspeed	1
Harsh acceleration	2
Crash events	0
Risk band	LOW

CITY RISK HEATMAP



Green = safe · Yellow = watch · Red = crash cluster

Boring where it should be, sharp where it matters.

MOBILE

- Flutter — Android + iOS single codebase
- Foreground service for trip capture
- On-device buffering + retry
- Battery-safe adaptive sampling

BACKEND

- Python + FastAPI (async ingest)
- PostgreSQL + TimescaleDB
- Redis (live score cache)
- Kafka / NATS event bus

AI / ML

- XGBoost risk classifier
- SHAP for feature attribution
- LLM (Claude / GPT) for coaching
- MLflow model registry

WEB & OPS

- Next.js insurer console
- Mapbox / OSM heatmap
- Docker + GitHub Actions CI
- OpenTelemetry + Grafana

Judge rides. Score reacts. Insurer sees it.

START
01

Judge taps **START RIDE**. App begins GPS + IMU capture on device.

RIDE
02

Simulated / real 5-min ride. Events stream to backend in real time.

DETECT
03

Engine flags **1** harsh brake and **2** overspeed events with severity.

SCORE
04

Rider Score = **84/100**. LLM explains the −6 delta with numbers.

REWARD
05

"Safe Rider" partial reward. Renewal benefit preview appears.

CONSOLE
06

Insurer dashboard updates — new data point lands on the city heatmap.

Sensor → AI → Score → Insurance action — the full loop in under 6 minutes.

No slideware, no smoke — the platform runs on stage.

WHY THIS WINS

Market, moat, and money.

220M+	~75%	₹90k Cr	< 3%
TWO-WHEELERS ON INDIAN ROADS	OF ROAD FATALITIES INVOLVE 2W*	MOTOR INSURANCE GWP (FY24)	CURRENTLY ON ANY UBI PRODUCT

VALUE CAPTURED

For riders	Cheaper cover for safe behaviour, real coaching, life-saving SOS.
For insurers	Lower loss ratio, better risk selection, stickier renewals, fraud defence.
For regulator	Data-driven road safety, aligned with IRDAI's UBI and telematics push.
For platform	SaaS + per-policy revenue + aggregated risk-data product for OEMs, RTOs.

*MoRTH road accident report, latest year available. Numbers used for context, not actuarial claims.

From hackathon MVP to pilot in 90 days.

1 **HACKATHON** — NOW
MVP: app + score + LLM explain + insurer console demo.

2 **PILOT** — 30 DAYS
500 riders, 1 insurer partner, IRDAI sandbox alignment.

3 **SCALE** — 60 DAYS
5k riders, refined risk model, closed-loop reward engine.

4 **PARTNER LAUNCH** — 90 DAYS
Co-branded product with insurer + OEM integrations.

BimaRide AI — Reward Safer Riding.